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WE ARE WHAT WE EAT: ANCIENT AGRICULTURE BETWEEN THE GREAT LAKES[1]

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INTRODUCTION: AN OLD STORY

NEARLY three millennia ago, between eastern Africa's Great Lakes, a long period during which food-producing communities settled largely environmentally distinct zones drew to a close with the arrival of Eastern Highlands Bantu-speaking farmers. Around 500 B.C. this earlier era of specialized foodways gave way to widespread contacts among the different communities, with Bantu-speaking peoples incorporating the technical agricultural knowledge of their predecessors in the region. In the centuries before A.D. 500, after more than 700 years of this complex Lakes Bantu synthesis, a period of consolidation set in. Lakes Bantu-speaking communities then spread fully through the region, possessed of a rich and diverse agricultural synthesis that supported population growth, shaped and responded to environmental changes and engendered unique regional variants.

These vital changes in food systems began in the Neolithic and drove the elaboration of the Early Iron Age. After the turn of the Christian Era, environmental change, population growth and settlement expansion all induced the creation and colonization of microenvironments best exploited by increasingly complex crop rotations and mixed farming using both wooden and iron implements. The broadly similar but areally differing foodways undergirded the growth of the regional cultural identities depicted in the famous corpus of Lakes oral histories. The agricultural past of the Kivu Rift provided the states in Rwanda, Buha and Kivu with the productive basis for struggles over their formation, organization and survival.[2] And, the peoples who lived between the Rift and Victoria Nyanza developed the

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[2] Jan Vansina, *L'évolution du royaume du Rwanda* (Brussels, 1960); J. Keith Rennie, 'The pre-colonial kingdom of Rwanda: a reinterpretation', *Transafrican Journal of History*, II (1972), 10–53.

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horticultural and pastoral base used during the eras recalled in oral traditions as the times of the AbaTembuzi and AbaCwezi in ancient Kitara.[3]

Previous work

Though not the first foray into the history of these early centuries, mine is the most thorough historical treatment of Great Lakes food systems in those times. Beyond some important overviews of the east African interior,[4] little detailed work on the early social history of the Great Lakes region as a whole has been published.[5] What is available has focused on either comparative linguistics or on archaeological approaches to the Early Iron Age.

Archaeologists correctly have focused first on questions of chronology and material cultural typologies,[6] only very rarely tackling the more slippery issues of economic and social change.[7] The best work demonstrates important

[3] Iris Berger, *Religion and Resistance: East African Kingdoms in the Precolonial Period* (Tervuren, 1981); Renee Tantala, 'The early history of Kitara in western Uganda: process models of religious and political change' (Ph.D. thesis, University of Wisconsin-Madison, 1989); and Bethwell A. Ogot, 'The Great Lakes', in D. T. Niane (ed.), *Africa from the Twelfth to the Sixteenth Century* (Paris, 1984), 498–524.

[4] John E. G. Sutton, 'East Africa before the seventh century', in G. Mokhtar (ed.), *Ancient Civilizations of Africa* (UNESCO General History of Africa, II) (Paris, 1981), 277–99; Roland Oliver and Brian Fagan, 'The emergence of Bantu Africa', in John D. Fage (ed.), *The Cambridge History of Africa*, (Cambridge, 1978), Vol. 2: *c. 500 B.C.–1050 A.D.*, 342–409; Michael Twaddle, 'Towards an early history of the East African interior', *History in Africa*, II (1975), 147–81; Christopher Ehret, 'The East African interior', in M. el-Fasi (ed.) and I. Hrbek (asst. ed.), *Africa from the Seventh to the Eleventh Century* (UNESCO General History of Africa, III) (Paris, 1988), 616–42.

[5] For the period 500 B.C. to A.D. 500 there is David L. Schoenbrun, 'Early history in Eastern Africa's Great Lakes Region: linguistic, ecological, and archaeological approaches. ca. 500 B.C. to ca. A.D. 1000' (Ph.D. thesis, University of California at Los Angeles, 1990). Although focused on periods after A.D. 1000 the following works are extremely suggestive of the new directions we seek to take here: David William Cohen, 'The face of contact: a model of a cultural and linguistic frontier in early Eastern Uganda', in Rainer Vossen and Marianne Bechhaus-Gerst (eds.), *Nilotic Studies: Proceedings of the International Symposium on Languages and History of the Nilotic Peoples, Cologne, January 4–6, 1982* (Berlin, 1983), 341–55; David S. Newbury, *Kings and Clans: Ijwi Island and the Lake Kivu Rift, 1780–1840* (Madison, 1991), 43–54.

[6] Peter R. Schmidt, *Historical Archaeology: A Structural Approach in an African Culture* (Westport, Conn., 1978); Francis Van Noten, 'The Early Iron Age in the Interlacustrine Region: the diffusion of iron technology', *Azania*, XIV (1979), 61–80; Peter R. Schmidt and S. Terry Childs, 'Innovation and industry during the Early Iron Age in East Africa: the KM2 and KM3 sites of northwest Tanzania', *African Archaeological Review*, III (1985), 53–94; Marie-Claude Van Grunderbeek, Emile Roche and Hugues Doutrelepont, *Le premier âge du fer au Rwanda et au Burundi: archéologie et environnement* (Brussels, 1983); Marie-Claude Van Grunderbeek, 'Essai d'étude typologique de céramique urewe de la région des collines au Burundi et Rwanda', *Azania*, XXIII (1988), 11–56; Jean Hiernaux and Emma Maquet, 'Cultures préhistoriques de l'âge des métaux au Ruanda-Urundi et au Kivu (Congo belge): Ie partie', *Bulletin des Séances de l'Académie Royale des Sciences Coloniales*, VI (1957), 1126–49; Jean Hiernaux and Emma Maquet, 'Cultures préhistoriques de l'âge des métaux au Ruanda-Urundi et au Kivu (Congo belge): IIème partie suivie de: deux sites archéologiques à briques en territoire Walikale (Kivu)', *Académie Royale des Sciences d'Outre-Mer, Mémoire No. 8*, Nouveau Série 10, 2 (1960), 1–88; Bernard Clist, 'A critical reappraisal of the chronological framework of the early Urewe Iron Age industry', *Muntu*, VI (1987), 35–62.

[7] Schmidt, *Historical Archaeology*, 286–94; Van Grunderbeek *et al.*, *Le premier âge*, 1983.

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connections between iron working sites, changes in farming practices, struggles over the levers of political power[8] and ecological crises induced by the depredations of smelters and smiths on local forests.[9] Though scholars often give to iron the central role in explaining these processes in Buhaya[10] or Rwanda and Burundi,[11] they fail to consider closely the specific crops grown and animals kept between 500 B.C. and A.D. 500 and the requirements those food systems made on labor, soils and tools. Though smelters and smiths living between the Great Lakes engineered considerable environmental change, food production more generally was a vital economic and social medium for their work.

Linguistically inclined historians begin to fill in this background by offering data on food history. Christopher Ehret has constructed an invaluable outline of processes of ethnic interaction, economic activity, social organization and religious systems for the entire eastern African interior.[12] He found that Central Sudanic and Southern Cushitic mixed agriculturalists influenced Bantu cultivators.[13] Christopher Wrigley has used comparable methods and data to focus more narrowly on economy and ethnicity. He has painted a broadly similar picture of multi-cultural interactions within the Lakes region during the Early Iron Age and established the early importance of cattle in Lakes societies.[14] Those probes into the region's early social history invite synthesis around a new, more comprehensive historical framework.

THE WAY BACK: INTRODUCING THE EARLIEST AGRICULTURALISTS

To describe the rise to prominence of Bantu-speaking societies that practiced mixed cropping and fishing economies beginning in the last millennium B.C., I build on four major sets of evidence. The first is evidence from pollen studies for a massive forest constriction in the western and southern portions of the region that began at very early dates.[15] The second set of data is

[8] John E. G. Sutton, *A Thousand Years of East Africa* (Nairobi, 1991), ch. 2. See especially chapters 7, 8 and 11 in Schmidt, *Historical Archaeology*. The constriction of forests and the development of a complex iron smelting technology in Rwanda, Burundi and Buhaya are beyond doubt. But we may question whether or not the links made in royal traditions between iron smelting sites, ruling dynasties and the powerful AbaCwezi patron spirits represent actual coincidence in the early Iron Age or result from later attempts by new dynasties to bolster legitimacy by claiming connections with sacred places.

[9] Van Grunderbeek *et al.*, *Le premier âge*; Noten, *Histoire archéologique*. In related work, Childs has located a possible source of friction between the iron workers and the potters of Buhaya in their competition over swamp clays with plastic and refractory qualities desired by both specialists; S. Terry Childs, 'Clay resource specialization in ancient Tanzania: implications for cultural process', in Charles Kolb (ed.), *Ceramic Ecology Revisited* (Oxford, 1988), 24.

[10] Schmidt, *Historical Archaeology*, 280.

[11] Van Grunderbeek *et al.*, *Le premier âge*, 42.

[12] Ehret, 'Between', 626–42.

[13] Ehret, *ibid.* 627–9. For a recent and far-reaching reconsideration of his earlier positions, see Christopher Ehret, 'The African Great Lakes Region in the Early Iron Age: shifting mosaics of cultural and economic interaction' (paper presented at the African Studies Association meeting, St Louis, 1991).

[14] Christopher C. Wrigley, 'Cattle and language between the Lakes', *Sprache und Geschichte in Afrika*, VIII (1987), 247–80.

[15] For full references and discussion, see David L. Schoenbrun, 'The contours of vegetation change and human agency in eastern Africa's Great Lakes region: ca. 2000 B.C. to ca. A.D. 1000', *Forest and Conservation History*, forthcoming (1993).

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historical linguistic evidence for a full range of agricultural and food collecting strategies in distinct environmental zones, employed by herders and farmers speaking Cushitic, Sudanic and Bantu languages contemporaneous with the earliest forest constriction. The third set of sources is archaeological evidence for industrial iron production in Buhaya, Rwanda and Burundi, broadly synchronous with the break-up of the proto-Great Lakes Bantu speech community around 500 B.C. These data imply both the production of agricultural surplus to support smelters and smiths and the existence of a sufficiently robust farming sector to create demand for iron hoes. Fourth, a further body of linguistic evidence deals with the spread of various Great Lakes Bantu languages across virtually the entire region, the rise of mixed agropastoral food systems and the breakdown of niche-specificity in choosing settlement sites – all this beginning late in the last millennium B.C. Constructing an integrated historical framework of explanations for these complex but organically interrelated processes makes up the core of this paper.

Changing forests[16]

The Lakes Bantu agricultural synthesis developed in clearings that opened up as the virgin forests of four millennia ago receded before human population growth and reduced rainfall. Palynology, the reconstruction of vegetation regimes through pollens recovered from stratified deposits, has revealed three major phases of vegetational changes across the western and northern parts of the Great Lakes region.[17] First, as early as 3000 B.C. in Kigezi, and not later than 1000 B.C. throughout the region, lower- and medium-altitude forests in valleys began to shrink significantly, leaving rises in grassland pollen percentages in stratified deposits. Together with a probable shift to a drier climate overall, a long-fallow, valley-bottom farming system probably contributed to the spread of grasslands in most areas. A forest-based, long-fallow technique may have 'simulated' the natural forests north-west of Victoria-Nyanza and on the lower slopes of Rwenzori,[18] making these early food systems difficult to detect.

Then, beginning as early as 2000 years ago in Rwanda and Burundi, and certainly by 1500 years ago elsewhere, secondary and degraded forests expanded at the expense of more mature forest communities. Herders and farmers, who lived probably on the hillsides or valley bottoms of the area, cleared forests by burning and cutting[19] and may have utilized the remaining

[16] The analytical and dating methods associated with palynology and paleolimnology are well covered in the literature. See David L. Schoenbrun, 'Treating an interdisciplinary allergy: methodological approaches to pollen studies for the historian of early Africa', *History in Africa*, XVIII (1991), 323–48.

[17] A full list of references to the literature on paleoenvironmental change in the Great Lakes Region is presented in Schoenbrun, 'Early history', 618–54. Data that underlie the summary given here are discussed fully in Schoenbrun, 'The contours'.

[18] See Schoenbrun, 'Early history', 70–91.

[19] Large concentrations of silted material in association with charcoal from Ahakagyezi supports this reconstruction. See Alan C. Hamilton, David Taylor and Joseph O. Vogel, 'Early forest clearance and environmental degradation in south-west Uganda', *Nature*, CCCXX (1986), 164.

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ridgetop forest refuges as managed food sources in times of famine and/or as ritual and burial sites. The older, long-fallow practices began to intensify throughout the region well before 1500 years ago, leading to a reduction of even climax secondary forest.[20]

By a thousand years ago a constellation of vegetation types similar to those found today dominated the Great Lakes region. Though climatic shift and anomaly undoubtedly shaped some of the environmental changes recorded in the pollen record, the array and timing of specific elements of vegetation change implicate human agents. Thus, the human imprint on today's highland environment along the Nile–Congo divide may have 5000-year-old roots.

Different tongues, different food systems: history from linguistics

Historical linguistics reveals the crops and animals, and something of the agricultural technology, with which herders, farmers, hunters, gatherers, fisher(wo)men, smiths and smelters contributed to the demise of the forests. Vocabularies from Lakes languages contain traces of their historical relationships, of their geographical spread and of the long term coexistence of very different speech communities in the Great Lakes region between 500 B.C. and A.D. 500. These historical processes of settlement and contact not only place people at the center of environmental change but reveal the power of historical linguistics to uncover details of how and why Lakes peoples so transformed their environment.

Historical relationships among Great Lakes Bantu[21] languages may be reconstructed from comparison between core vocabulary lists[22] that allow

[20] Schoenbrun, 'Early history', 91–4.

[21] Schoenbrun, 'Early history,' ch. 5, presents the career of nomenclature for different sub-units of Great Lakes Bantu. I follow the spirit of the conference on 'the ancient civilisation of the peoples of the Great Lakes' where 'Great Lakes' or 'Grands Lacs' was substituted for 'Interlacustrine'. See *La civilisation ancienne des peuples des grands lacs* (Colloque de Bujumbura, 4–10 septembre, 1979) (Paris and Bujumbura, 1981), 7. To the historical linguist, Great Lakes Bantu is a sub-branch of the Eastern Highlands group of the Niger–Congo family of African languages. The use of the term 'Eastern Highlands Bantu' refers to Bernd Heine's 'Ostochland' group and reflects a genetic classification. The term 'Eastern Bantu', where used below, refers to a referential classification. Unfortunately, many of the people who spoke proto-Eastern Highlands dialects probably did not live in highland environments! Bernd Heine, 'Zur gentische Gliederung der Bantu-sprachen', *Afrika und Übersee*, LVI (1973), 164–85; and Bernd Heine, H. Hoff and Rainer Vossen, 'Neuere ergebnisse zur Territorialgeschichte der Bantu', in Wilhelm Möhlig, Franz Rottland and Bernd Heine (eds), *Zur Sprachegeschichte und Ethnohistorie in Afrika* (Berlin, 1977), 57–72. The use of these terms does not imply agreement with the formulation 'Western Bantu' as presented by Jan Vansina, 'Western Bantu Expansion', *J. Afr. Hist.*, XXV (1984), 129–45. The great time depth of the proposed binary split makes suspect any classificatory decisions based solely on lexicostatistical study. But see Yvonne Bastin, André Coupez and Bernard de Halleux, 'Classification lexicostatistique des langues bantoues (214 relevés)', *Bulletin des Séances de Académie Royale des Sciences d'Outre-Mer*, XXVII (1983), 173–99. We await the next effort by the team at Tervuren, in association with Michael Mann, which will include some 345 Bantu languages.

[22] Material for these lists comes from my field collections, those of Derek Nurse, Gerard Philippson and their students at UDSM and other published and unpublished sources. See Schoenbrun, 'Early history', 364–411, 422–49.

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provisional establishment of regular sound correspondences and the recognition of cognates. For example, in KiHa (a West Highlands language spoken in north-west Tanzania, see Fig. 1 and Map 1) the stem *-kùùra* means 'to uproot a tuber', and in RuKoonzo (a Rwenzori language spoken on Uganda's side of the Rwenzori mountains, see Fig. 1 and Map 1) *-kùùrha* means 'to uproot beans and groundnuts'. Since both stems cover the same semantic territory with corresponding sounds, they are cognates; they are inherited items in KiHa and RuKoonzo. By counting rates of cognation in core vocabulary, for each pairing of the languages under study, a preliminary sub-grouping of Great Lakes Bantu can then be adduced. Higher percentages show more recent divergence, while lower numbers show earlier divergence. KiHa and RuKoonzo have a 59 per cent cognation rate; they share 59 out of the 100 core vocabulary items. But KiHa and KinyaRwaanda have a 83 per cent cognation rate. All three tongues are related, but RuKoonzo began to diverge from its ancestral speech community, proto-Rwenzori, long before KiHa and KinyaRwaanda began to diverge from theirs, proto-West Highlands.

These ancestral or proto-languages represent the period of several centuries (or longer) during which dialects were still mutually intelligible. Thus, the proto-West Highlands and proto-Rwenzori periods are only now drawing to a close, because many speakers of RuKoonzo and LuNande (Rwenzori) still understand each other clearly, and many speakers of KiHa, KiRundi, KinyaRwaanda and KiVinza (West Highlands) understand each other as well. However, a RuKoonzo speaker and a KiHa speaker would understand each other only dimly, as their 59 per cent cognate score betokens a long departed proto-Western Lakes community.[23]

Just how long departed is the proto-Western Lakes community? Chronological dimensions of Lakes language history come also from the evidence of lexicostatistics. Glottochronology, a subset of lexicostatistics, uses the empirical finding that random change in basic vocabulary tends, over periods of centuries, to accumulate at a regular pace, and thus differences in cognate percentages can be given extremely rough chronological value.[24] About 16 out of 100 items will be replaced each 500 years, either by semantic shift, by morphological analogy or by borrowing from another language.[25] Thus, using the rate given above, KiHa and KinyaRwaanda ceased easily to be mutually intelligible about 500 years ago, and RuKoonzo's ancestral speech

[23] Only the Western Lakes and West Nyanza branches of Great Lakes Bantu are discussed in this essay. Greater Luhyia, East Nyanza and Pre-RuGungu are not pursued.

[24] For generally favorable views on this vexed question see Morris Swadesh, 'Towards greater accuracy in lexicostatistical dating', *International Journal of American Linguistics*, XXI (1955), 121–37; Raimo Anttila, *Historical and Comparative Linguistics* (Amsterdam, 1989), 396–8; and Christopher Ehret, 'Language change and the material correlates of language and ethnic shift', *Antiquity*, LXI (1988), 566–9. For unfavorable views see Theodora Bynon, *Historical Linguistics* (Cambridge, 1977), 266–72; and Hendrik Birnbaum, *Linguistic Reconstruction, its Potential and Limitations in New Perspective* (Washington, 1978), 17.

[25] All such 'replacements' are innovations and represent natural processes of language change that occur in both literate and non-literate contexts. See Jan Vansina, *Paths in the Rainforests : Toward a History of Political Tradition in Equatorial Africa* (Madison, 1990), 9–16; Ehret, 'Language change', 564–6; Schoenbrun, 'Early history', 103–31.

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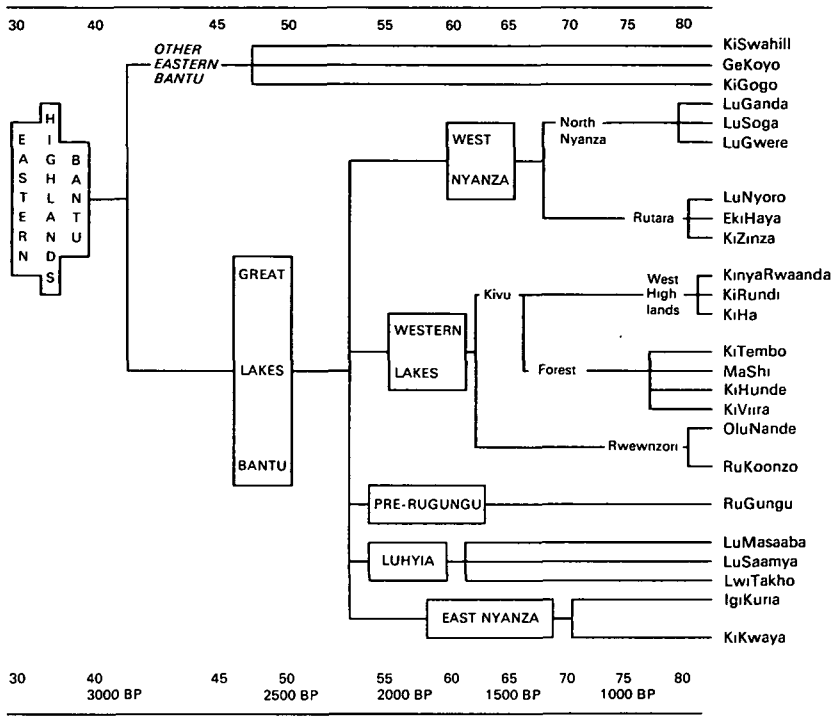


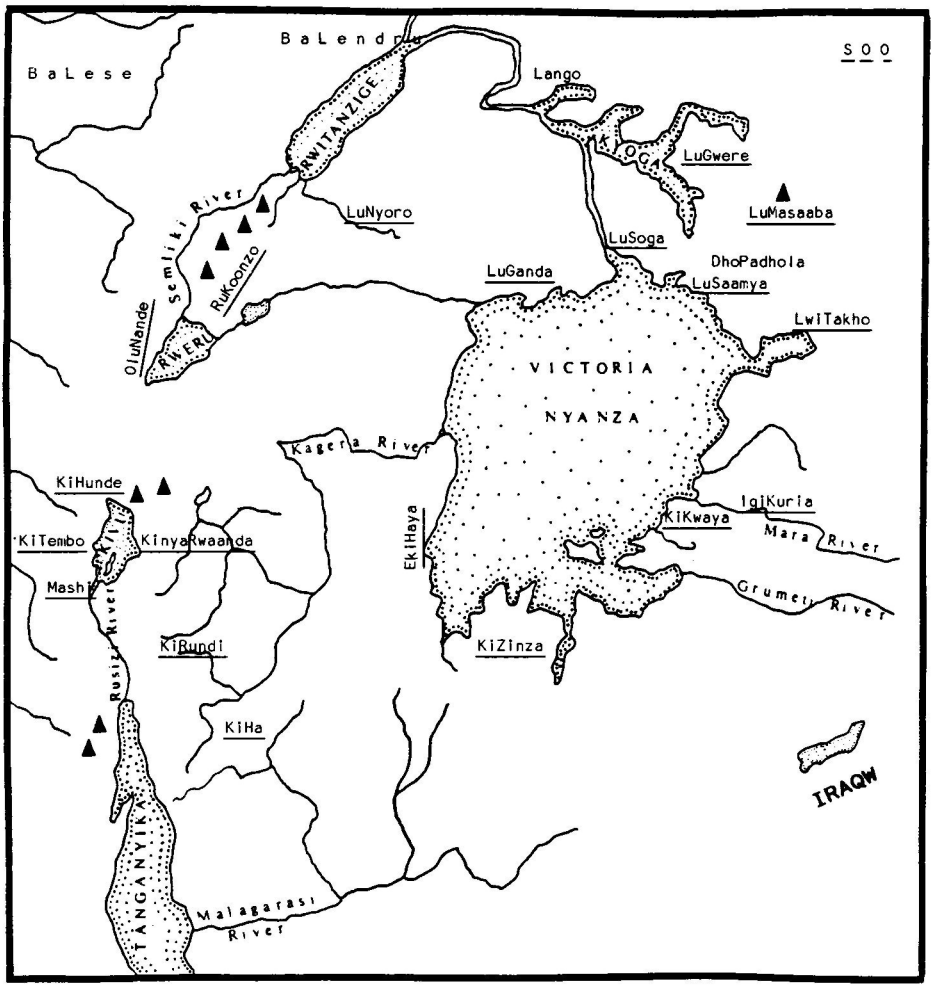
Fig. 1. The genetic position of Great Lakes Bantu and its descendants. Per cent cognation out of a 100-word list. Glottochronology reckoned at a shared retention rate of 86 per cent per 500 years. The dateline should be logarithmic but is not; it corresponds to the protolanguages in order to show their age.

community, Rwenzori and Kivu (out of which both KiHa and Kinya-Rwaanda were to grow) lost easy mutual intelligibility over 1500 years ago. Fig. 1 represents the historical relationships between several intermediate groups of Great Lakes Bantu and gives a rough chronology for their development.

Applying the principle of least moves,[26] I can suggest the directions taken as Lakes Bantu speech diversified and spread across the region. Similar to Occam's razor, I begin with modern languages and map their geographical extent (Map 1) and attribute equal strength first to modern tongues, and then to their ancestral speech communities as they must have moved across the geographical plane in order to account for the current distribution. Thus, in Fig. 1, since the bundle of modern languages constituting the North Nyanza branch of West Nyanza Bantu centers on the north-west corner of Victoria Nyanza, and because the corresponding group of languages making up Rutara centers on the area south of the mouth of the Kagera river (Map 1), these all must have developed out of a zone midway between these two areas; the West Nyanza territories lay then around the Victoria Nyanza shore and inland along the Kagera river (Maps 3C and 4B). The question of how

[26] Edward Sapir [1916], 'Time perspective in aboriginal American culture: a study in method', in David G. Mandelbaum (ed.), *Edward Sapir: Selected Writings in Language, Culture, and Personality* (Berkeley, 1985), 410–25.

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Map 1. Modern linguistic geography: the Great Lakes Region. (Key: LuGanda, Great Lakes Bantu; BaLese, Central Sudanic; *SOO*, Kuliak [Eastern Sudanic]; IRAQW, Southern Cushitic; Lango, Western Nilotic.)

languages spread is at once historical and sociological.[27] The principle of least moves is entirely instrumental and merely approximates the historical regions in which different Lakes proto-linguistic identities in fact would have emerged. Social, environmental and economic factors also shaped the spread of Lakes Bantu speech, and I shall touch on all of these in the remainder of the essay. But the early spread of proto-Great Lakes Bantu certainly was facilitated by settlement along the lakeshores and riverbanks and by the use of canoes.[28]

[27] See the collection of papers in Robert L. Cooper (ed.), *Language Spread: Studies in Diffusion and Social Change* (Bloomington, Indiana, 1982), especially 5–62.

[28] Ehret, 'Linguistic inferences about early Bantu history', in Christopher Ehret and Merrick Posnansky (eds.), *The Archaeological and Linguistic Reconstruction of African History* (Berkeley, 1982), 59–61; Vansina, *Paths*, 55–6.

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Although today the region is dominated by Bantu-speakers, members of other groups are found in adjacent areas (Map 1). Their present remoteness from the central Great Lakes area conceals the important role their ancestors played in the early history of the region. Loanwords in Lakes Bantu languages from Central Sudanic, Sog Eastern Sudanic and Tale Southern Cushitic mark partially their contributions to the region's agricultural and cultural history. They are given in the Appendix.[29] Identifying loans in Great Lakes Bantu languages depends upon a thorough understanding of the phonological history of likely source languages as well as for Lakes Bantu languages, attained in the search for cognates. The value of loans lies in their indicating who was involved in the contact and what was transferred. Such borrowings form a major historical source.

But the study of the internal development of lexicon concerned with food production practices also sheds light on historical processes of increasing complexity. For example, by tracing the speciation of vocabulary items – both loans and internal innovations – the historian may reason about the timing and the choices taken by some Lakes Bantu-speakers as, for instance, they came gradually to incorporate livestock into their productive economy. Some items were common to Great Lakes Bantu while other words were coined later by West Nyanza and Western Lakes speakers. This fact reveals both that, in the period between Great Lakes Bantu and West Nyanza and Western Lakes, livestock practices changed and that they did so differently in the two descendant groups. Thus, the systematic study of cultural vocabulary and loanwords connected with food production reveals the elaboration of different foodways and the extraordinarily diverse cultural contexts in which Lakes farmers and herders lived.

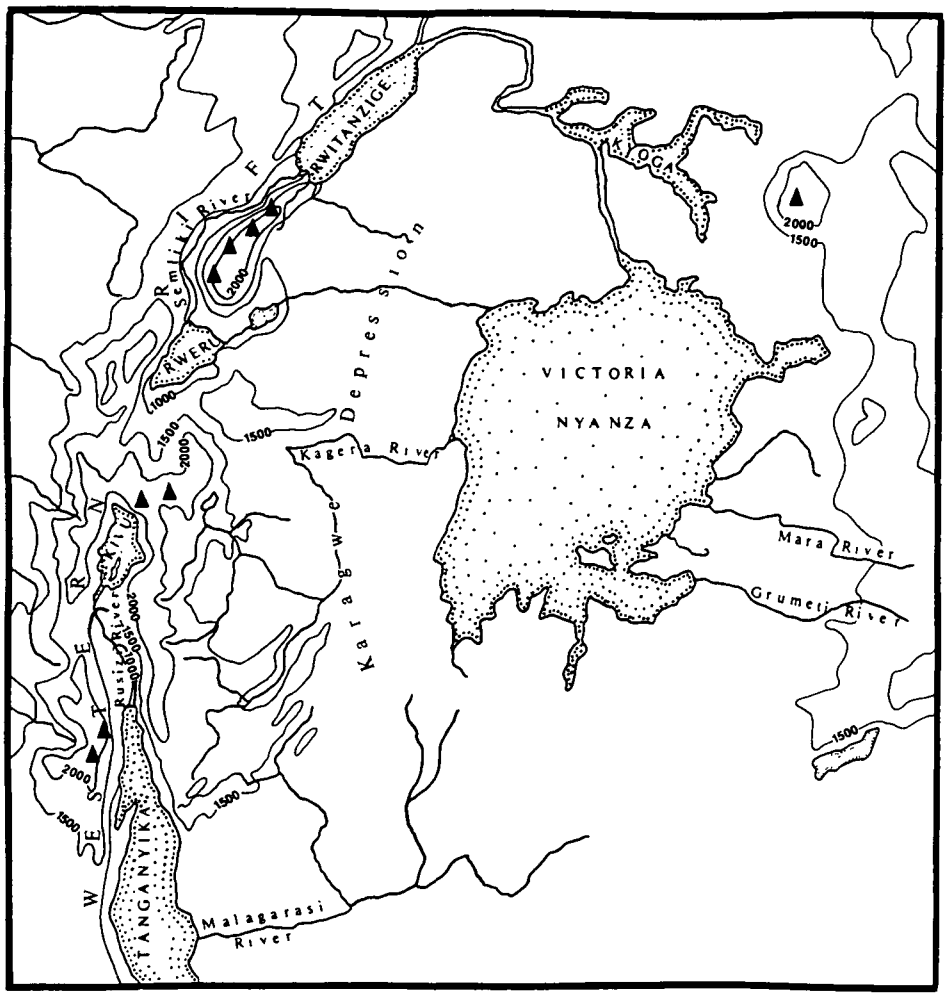
FOOD AND SETTLEMENT: ENVIRONMENTAL SPECIFICITY BEFORE *c.* 500 B.C.

In ancient times speakers of two distinct branches of Nilo-Saharan and an arm of the Southern Cushitic language communities[30] lived in the western portion of the Great Lakes region. But today these same areas are dominated by speakers of Great Lakes Bantu. All the non-Bantu-speaking groups need to have been present in parts of the Lakes region prior to the arrival of Eastern Highlands Bantu or the development of Great Lakes Bantu speech in order to account for the presence of loans in the latter communities (Appendix), so I begin with them. A focus on the Bantu side of the story is unavoidable in the lack of comparable linguistic evidence from the extinct Central and Eastern Sudanic and Tale Southern Cushitic vocabularies. Just what they learned in return from the Eastern Highlands and Great Lakes Bantu remains a mystery.

[29] When mentioned in the text the word's corresponding number will be given in parentheses.

[30] Christopher Ehret, 'Cattle-keeping and milking in Eastern and Southern African History: the linguistic evidence', *J. Afr. Hist.*, VIII (1967), 1–18; *idem*, 'Sheep and Central Sudanic peoples in Southern Africa', *J. Afr. Hist.*, IX (1968), 213–22; Derek Nurse, 'Extinct Southern Cushitic communities in East Africa', in Marianne Bechhaus-Gerst and J. Serzisko (eds.), *Cushitic-Omotic : Papers from the International Symposium on Cushitic and Omotic Languages, Cologne, 6–9 January, 1986* (Hamburg, 1988), 93–104; Schoenbrun, 'Early history', chs. 5, 6.

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Map 2. (*a*) Altitude (in meters).

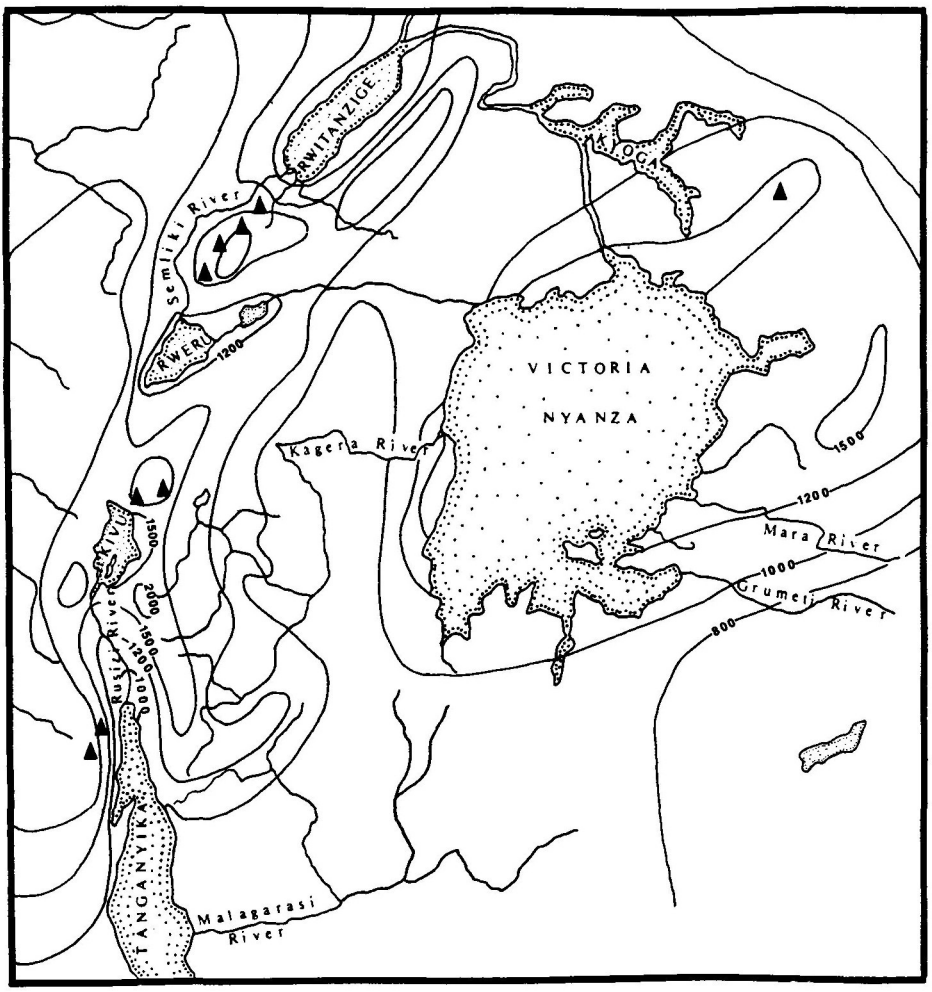
Central Sudanic

In the centuries before 2500 years ago several Central Sudanic-speaking societies resided at the western side of the Great Lakes region.[31] Though neighbors of the proto-Great Lakes Bantu, the basic building blocks of their economies were quite distinct. They raised and milked cattle and grew sorghum and possibly *Pennisetum* millet.[32] Sorghum and *Pennisetum* pro-

[31] This and subsequent discussions of the chronology and location of settlement in the Great Lakes region based on historical linguistic criteria are bolstered by correlations with a sometimes uneven archaeological record of ceramic traditions. The methodological focus of this exercise often deflects attention from the history of economic and social process, though precisely economic and social factors are implicated in the development, dispersal and promotion of a ceramic industry. See Schoenbrun, 'Early history', 267–90, for ideas about correlation both of the historical linguistic record and regional pottery traditions.

[32] For comparative linguistic evidence see Christopher Ehret, 'Patterns of Bantu and Central Sudanic settlement in Central and Southern Africa (ca. 1000 B.C.–500 A.D.)',

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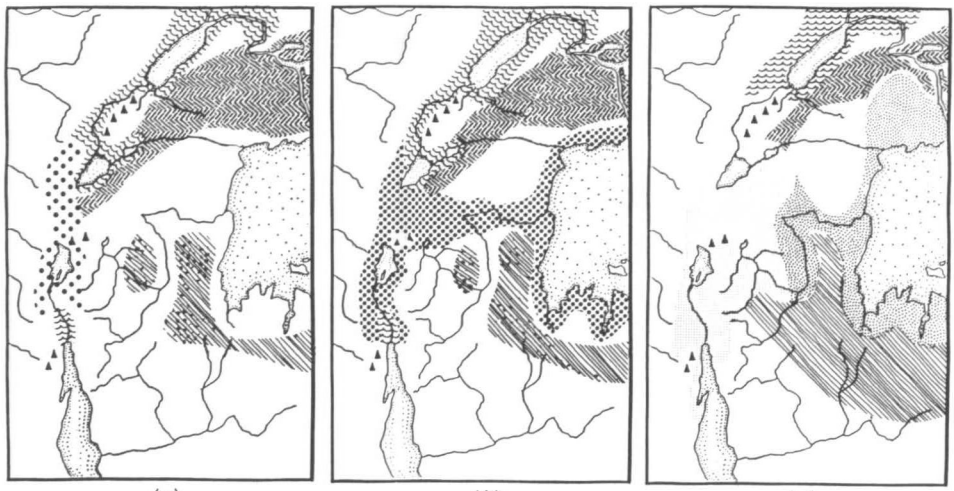


Map 2. (*b*) Rainfall (in millimeters).

Transafrican Journal of History, III (1973), 1–71; *idem*, 'Agricultural history in Central and Southern Africa, c. 1000 B.C. to A.D. 500', *Transafrican Journal of History*, IV (1974), 4ff; and *idem*, 'Proto-Central Sudanic reconstructions' (n.p., 1986). For more work on Central Sudanic see Didier Demolin, 'Some problems of phonological reconstruction in Central Sudanic', *Belgian Journal of Linguistics*, III (1988), 53–96; Didier L. Goyvaerts (ed.), *Language and History in Central Africa* (Antwerp, 1986); and M. Lionel Bender, 'Central Sudanic segmental and lexical reconstruction', *Afrikanistische Arbeitspapiere*, XXIX (1992), 5–61.

Sorghum and *Pennisetum* both thrive in lower rainfall regimes with much less atmospheric moisture than required for the yam or oil palm. Sorghum cannot tolerate frost but can briefly withstand waterlogged soils without suffering significant loss of yield. See J. W. Purseglove, *Tropical Crops, Monocotyledons* (Harlow, Essex, 1972), 269–71. In eastern Africa, *Pennisetum* requires a minimum average rainfall of 350–400 mm; Food and Agriculture Organization, *Traditional Food Plants: A Resource Book for Promoting the Exploitation and Consumption of Food Plants in Arid, Semi-Arid, and Sub-Humid Lands of Eastern Africa* (Rome, 1988), 472. *Pennisetum* sp. needs a more evenly distributed rainfall regime than does Sorghum, although it will grow with as little as 250 mm of rain annually; Purseglove, *Monocotyledons*, 206. In eastern Africa, *Pennisetum*

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Map 3. (*a*) Before 800 B.C.; (*b*) from 500 B.C. to A.D. 100; (*c*) from A.D. 100 to A.D. 500. (Key: , Eastern Highlands Bantu; , Tale Southern Cushitic; , Sog Eastern Sudanic; , Central Sudanic; , Great Lakes Bantu; , Western Lakes Bantu; , West Nyanza Bantu.

vided versatility to Central Sudanian farmers when choosing cultivation sites that makes it difficult to draw a line on a map defining clearly where they lived early in the last millennium B.C. But the fact that eastern African farmers today rarely plant *Pennisetum* above 1200 meters limits the probable range of settlement choices for those Central Sudanians farming it to the Ruzizi River valley at the northern end of Lake Tanganyika and to the east and north-east of the Rwenzori in the lower reaches of the Semliki (compare Maps 2, 3 and 4). Other locations where Central Sudanians lived could have been in the lower altitudes in the north-western Kagera depression and toward the shores of Lake Rwitanzige in modern-day BuTooro and BuNyoro (Map 3A). Additionally, in order for Great Lakes Bantu farmers to have borrowed words for crops and animals from Central Sudanic farmers (5–9 in Appendix), the Sudanians had to have been nearby or even in territories occupied by Lakes Bantu speakers.

Central Sudanians also bred livestock and exploited them for such secondary products as milk. A host of nouns in Central Sudanic for stock and for milk products attests to this.[33] And a single verb, 'to exchange (animal)', reveals directly a means for the transfer into Great Lakes Bantu of Central Sudanic words. As a productive source of food and potential trade item and as a unique form of reproductive and self-transporting wealth, livestock raising exists within a complex human ecological net that has undoubtedly

requires high temperatures during the day and cool ones at night to mature properly. The mean minima and maxima should be 18 °C and 27 °C respectively. The crop is rarely found above 1200 m, FAO, *Traditional Food Plants*, 396–7.

[33] The authorities here are Ehret, Demolin and Goyvaerts (see note 32). But work undertaken in Zaire by Mary McMaster of Castleton State University, and soon to be underway by Kairn Klieman of UCLA, should add greatly to our knowledge.

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changed greatly over time.[34] But, assuming relatively low population densities before 500 B.C., Central Sudanic herders probably grazed their animals in areas without forest. Early Central Sudanic herders thus would have found pasture either high above the forest belts (above about 2800 m) or in the drier corridor north-west of the Kagera river. Cattle trypanosomiasis and tick-borne diseases might have influenced patterns of access to pasture.[35] While grain farmers likely restricted themselves to the lower altitudes in this area, some seasonal movement in search of dry season flush, available at higher altitudes, probably occurred.[36] Such transhumance provided at least an intermittent form of contact between Bantu- and Central Sudanic-speaking peoples, as small groups of herders made their way through the more open forest edges, where casual contact with the tsetse fly may have created a resistance to trypanosomiasis.[37] Though very likely the same areas that root-cropping proto-Great Lakes Bantu farmed, enjoying abundant light and forming natural discrete ecotones, Central Sudanic herders probably settled thinly at the upper margins of this band of territory. To the east, across the Karagwe Depression, lay another important set of communities who spoke various dialects of Eastern Sudanic.

Sog Eastern Sudanic[38]

Central Sudanians were the main players, especially in the earliest eras of the history of the farther western portion of our region, but they were not the only players.[39] Loanword evidence (15–24 in Appendix) reveals an ancient

[34] Douglas H. Johnson and David M. Anderson (eds.), *The Ecology of Survival: Case Studies from Northeast African History* (London and Boulder, CO, 1988); Richard Waller, 'Ecology, migration, and expansion in East Africa', *African Affairs*, LXXXIV (1985), 347–70.

[35] Ford, pointing out the speculative nature of such considerations, concludes that modern distributions of fly fail to indicate ancient patterns of tsetse infestation, but that 'a vegetational and faunal continuity existed that would have allowed *fusca* group tsetses and their hosts to have spread along the forest edges associated with the highlands of both the Western and Eastern Rifts...'; John Ford, *The Role of the Trypanosomiases in African Ecology: A Study of the Tsetse Fly Problem* (Oxford, 1971), 121. For the sake of simplicity I assume, with Ford, that areas of more open grasslands above about 1500 m were relatively safe for livestock; Ford, *ibid.* 5. I also assume that these same areas might have harbored tick-borne 'east coast fever'. Comparative linguistic data might help to prove the existence of these pests during the early periods of history. A root **-gUpe* meaning 'cattle tick' in proto-Great Lakes Bantu is the strongest evidence so far. A shift in meaning from the general term for 'tick' to that of 'cattle tick' seems to have occurred during the proto-Great Lakes period.

[36] Neville Dyson-Hudson, *Karimojong Politics* (Oxford, 1966), 39; John Weatherby, 'Inter-tribal warfare on Mount Elgon in the 19th and 20th centuries', *Uganda Journal*, XXVI (1962), 200–12.

[37] Ford, *The Role*, 86–90 referenced in James Giblin, 'Trypanosomiasis control in African history: an evaded issue', *J. Afr. Hist.*, XXXI (1990), 63.

[38] The name 'Sog' is the Kuliak word for 'mountain' and is a common loan word in Great Lakes Bantu. See root 23 in Appendix.

[39] We are ignoring the question of the presence and role of specialized gatherer-hunters in the Great Lakes region. Since no linguistic evidence yet exists which might be of use in discussing exclusively gathering and hunting peoples the lacuna is justifiable if lamentable. Later efforts at reconstructing vocabulary for hunting, wild food collecting and fishing should reveal the rich contributions these activities made to Great Lakes nutrition.

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Map 4. Linguistic geography: early first millennium A.D. (*a*) Western Lakes area; (*b*) West Nyanza area. (Key: , Western Lakes Bantu; , Tale Southern Cushitic; , Central Sudanic; , West Nyanza Bantu; , Sog Eastern Sudanic.

speech community no longer in existence but related to the Proto-Kuliak branch of Eastern Sudanic from which descended the modern Kuliak languages, spoken today mainly in north-eastern Uganda (Map 1).[40]

Sog Eastern Sudanic loans entered Lakes Bantu speech over nearly the

[40] I owe to Christopher Ehret the original observation that several Nilo-Saharan loans in Great Lakes Bantu could not be derived from Nilotic or Central Sudanic sources. Based on linguistic geography and its correlation with archaeological sequences, John Sutton long ago hinted at an ancient Nilo-Saharan presence in the Lakes region; John E. G. Sutton, 'The Aquatic Civilization of Middle Africa', *J. Afr. Hist.*, XV (1974), 537. For a collection of reconstructed agricultural terms in Proto-Kuliak see Schoenbrun, 'Early history', 570–1, where all items come from Bernd Heine, *The Kuliak Languages of Eastern Uganda* (Nairobi, 1976), 73–9; Christopher Ehret, 'The classification of Kuliak', in Thilo C. Schadeberg and M. Lionel Bender (eds.), *Nilo-Saharan : Proceedings of the First Nilo-Saharan Linguistics Colloquium* (Dordrecht, 1981), 269–89; and *idem*, 'Revising Proto-Kuliak', *Afrika und Übersee*, LXIV (1981), 81–100.

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whole millennium, from 500 B.C. to A.D. 500. Proto-Great Lakes Bantu speakers learned words for 'cow' and 'porridge' from the Sog, implying that cultural interaction between the two groups revolved around livestock and grain. The remainder of this evidence, however, substantiates three separate areas and periods of contact between different Great Lakes Bantu societies and these groups of Eastern Sudanic-speaking people: the first, in the Luhyia territory from around 2000 years ago; the second, in the East Nyanza area due south of the Winam Gulf beginning in the centuries before *c.* A.D. 200 or 300; and, the third, at about the same time, in the lands of proto-West Nyanza settlement (Maps 3C and 4B).

Words for food production in Kuliak reveal that, around the homesteads in the north and north-eastern sections of the Great Lakes region, Sog farmers practiced an agriculture vaguely reminiscent of their distant cousins, the Central Sudanians. They grew sorghum and beans but seem to have been involved equally in raising and milking livestock. They bred both small stock and cattle, as the reconstructed Kuliak breeding taxonomies imply. And we might imagine both the southern Kuliak and the Sog Eastern Sudanic peoples to have inhabited ecological zones very similar to the newly arrived East Nyanzan and Luhyia communities – the lower, more open stretches of grassland and secondary forest in the north-eastern sector of the Great Lakes region.[41]

The southerly extensions of Kuliak-related people along the east and west of Victoria Nyanza seem to have disappeared in the last centuries of the first millennium A.D.: no new Sog Sudanic loans turn up in the Bantu-speaking communities who descended from East Nyanza and West Nyanza. At least north-west of Victoria-Nyanza, before A.D. 500, the presumably more densely settled West Nyanza Bantu communities, supported by fishing, hunting, root-cropping, cereal-raising and livestock-herding, may have impeded the Sog people's access to critical resources such as fertile soils, pasture, water and salt. Assimilation into Bantu-speaking societies would have been one response for the Sog speakers, bringing with them their knowledge and terminology for grain farming and cattle.

Tale Southern Cushitic

If the important role of Central Sudanic societies in the earliest eras of the history of greater eastern Africa has been disputed,[42] and that of the Eastern Sudanians largely unrecognized, then the place of various Southern Cushitic-speaking communities has received considerably more sympathetic, and careful scholarly attention.[43] Among other contributions, their herding and

[41] The southern Kuliak may have been responsible for the earliest evidence of forest clearance in the Pilkington Bay pollen core. Their apparent reliance on sorghum as the main cereal crop nicely supports the interpretation of a long-fallow, valley-bottom farming system involved in the clearance of lower forests; Schoenbrun, 'Early history', 90–1. Sorghum is resistant to the waterlogging common in swales.

[42] David William Cohen, *The Historical Tradition of Busoga: Mukama and Kintu* (Oxford, 1972), 77ff; David W. Phillipson, *The Later Prehistory of Eastern and Southern Africa* (London, 1977), 219ff.

[43] Ehret's work furnishes the basis for the following review of Southern Cushitic agricultural practice, especially Christopher Ehret, *The Historical Reconstruction of Southern Cushitic Phonology and Vocabulary* (Berlin, 1980). For considerations of the role of Southern Cushitic-speakers elsewhere in eastern African history see also Stanley

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farming expertise developed in the midst of the earliest food producing communities between the eastern Rift valley and the coast.[44] But here we are concerned only with those communities living in the Great Lakes area, called now Tale Southern Cushites.[45] These herders and farmers settled areas both to the west and north-east of Victoria Nyanza (Map 3). And in order to explain their loans found in Great Lakes Bantu (25–30 in Appendix), Cushites had to have been already living there when Great Lakes Bantu communities moved in towards the end of the last millennium B.C. They were a major source of cattle-keeping knowledge for the early Bantu-speaking communities, even though the majority of borrowings belongs to later sub-groupings of Great Lakes Bantu.[46] Much of the following discussion will draw upon features of the food systems inherited from their older Southern Cushitic ancestors or from their more direct ancestral speech community, the Rift Southern Cushites.[47]

Settled in the southern parts of the region throughout the last millennium B.C., the bearers of Tale Southern Cushitic culture shaped Lakes history profoundly. They raised crops and animals broadly similar in range to those of the Eastern and Central Sudanians; probably including a couple of varieties of *Sorghum bicolor* and, possibly, *Eleusine coracana*.[48] However, pastoralism held a more central position compared to its role in the Sudanic agriculture. The Southern Cushite food system centered on the breeding, herding, milking and bleeding of cattle.[49] A word for 'red and white head of cattle', in proto-Rift,[50] suggests that cattle were important enough to Rift

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Ambrose, 'Archaeology and linguistic reconstructions of history in East Africa', in Ehret and Posnansky (eds.), *The Archaeological*, 140–2; Christopher Ehret, *Southern Nilotic History* (Chicago, 1971), 37–9, 48–51, 55–60, 110–27; *idem*, *Ethiopians and East Africans: The Problem of Contacts* (Nairobi, 1974), 7–44; Peter T. Robertshaw and David P. Collett, 'A new framework for the study of early pastoral communities in East Africa', *J. Afr. Hist.*, XXIV (1983), 289–301.

[44] There were several different communities of Southern Cushitic-speaking societies in greater eastern Africa. Christopher Ehret and Derek Nurse, 'The Taita Cushites', *Sprache und Geschichte in Afrika*, III (1981), 125–68; Derek Nurse, 'Reconstruction of Dahalo history through evidence from loanwords', *Sprache und Geschichte in Afrika*, VII (1986), 267–305; and Nurse, 'Extinct', 96, 100.

[45] The name replaces Ehret's 'Nyanza Southern Cushites', Ehret, *Ethiopians*, 24–7, in favor of a suitably Cushitic word, but it follows his and Nurse's conception of this linguistic grouping. Tale Southern Cushitic belonged to the West Rift branch of Southern Cushitic (as for example, does present-day Iraqw of north-central Tanzania).

[46] A quite detailed history of these contacts emerges from loanword studies; see Schoenbrun, 'Early history', 190–251, and Ehret, 'The African Great Lakes', 17–26.

[47] Ehret, *Historical Reconstruction*, 132.

[48] *E. coracana* is a very durable crop. It is extremely drought-resistant and may store seeds on the head for up to ten years, though viability for planting drops 50 per cent after two years. It thrives in free-draining soils with steady moisture levels. It does best between 18 °C and 27 °C, from sea-level to 2400 m. In Uganda it requires 900–1250 mm of rainfall and dislikes waterlogging; FAO, *Traditional Plant Foods*, 266–7.

[49] Proto-Rift Southern Cushitic vocabulary, and presumably Tale Southern Cushitic vocabulary, possessed detailed sets of terms for livestock breeding taxonomies, secondary products and probably a number of terms for specific cattle colors. See Ehret, *Historical Reconstruction*, 338–84.

[50] See Ehret, *Historical Reconstruction*, 347. This noun turns up as a loan in Rutara Bantu, a descendant of proto-West Nyanza Bantu, after A.D. 800 or 900.

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Southern Cushites to create new fields of meaning for keeping track of animals in large herds and for naming cows with unique coloration.

Southern Cushites, oriented as they were towards livestock, probably used land more extensively than either their Great Lakes Bantu or Sudanic neighbors. They may also have settled more thinly across the landscape than their neighbors in order to exploit the last available niche for food producers: the long-grass savannas along the southern portion of the Great Lakes region and to the south and east of Mount Elgon.[51] Additionally, Great Lakes Bantu communities borrowed two words naming the savanna-dwelling lion (28) and bush pig (29), indicating both the novelty to Bantu speakers of dry woody grassland environments and that the savanna-forest ecotone was one of the locations for their interactions with Tale Southern Cushites.

All three major language communities remained intensively in contact up to the beginning of the Christian Era, in the territory stretching south from the elbow of the Kagera River to the headwaters of the greater Malagarasi river (Maps 1, 4A and 4B). Equal access to desired resources, rather than competition for pasture or fertile soils, likely characterized these cross-cultural encounters. Yet, this sort of balance, based on the environmental specificity of the three agricultural profiles, had already begun to break down in the first centuries A.D. Almost without doubt, the spread of Bantu-speakers upset that balance. But, initially anyway, they were part of that equilibrium.

BEYOND NICHE SPECIFICITY: THE GREAT LAKES BANTU AGRICULTURAL SYNTHESIS, 500 B.C. TO A.D. 500

Although most historians and archaeologists implicitly have dated the development of agriculture to the Early Iron Age, *c.* 500 B.C. to *c.* A.D. 500, all the traditions of food production discussed so far from linguistic evidence developed before that time. What changed after 500 B.C. was the achievement by Great Lakes Bantu peoples of a mixed agricultural synthesis that incorporated virtually all of the food productive expertise developed in Sudanic and Cushitic homesteads. This new agriculture enabled them to break down the earlier environmental specificity and to make the world of the Lakes region a Bantu-speaking one. What began as a process of cultural interaction among speakers of diverse languages in the last millennium B.C. grew into an expansion of a single, Bantu speech and culture, the story of which gives life to the pots, tuyères and furnaces that characterize the Early Iron Age itself. This large-scale historical process took shape differently across the varied environments of the region and had one of its own roots in the inherited tradition of Eastern Highlands food production practiced and

[51] This sort of observation has been made most recently by Ehret, 'The African Great Lakes', 52, and for pastoralists in the Kenya Rift by Ambrose, 'Archaeology', 140–1. Later work provided an even better 'fit' between material cultural traditions, environmental zones and Southern Cushitic sub-groups, though the authors did not intend this as a conclusion. Compare Ehret's historical linguistic discussion of Southern Cushitic in *Ethiopians*, 7–31, with the distributional dimension of the four different traditions recognized in the new classification of Pastoral Neolithic pottery assemblages by Robertshaw and Collett, 'A new framework', 290 and 292.

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elaborated upon in Great Lakes Bantu fields and pastures. As the only continuous tradition in the region, Lakes Bantu history deserves an equally uninterrupted account from earliest times to A.D. 500. The remainder of the paper tells this story.

Even a cursory look at the multiple levels of the outline classification of Great Lakes Bantu, given in Fig. 1, generates respect for the antiquity of this community of speakers. Its most distantly related languages share about 40 per cent of their basic vocabulary, indicating that speakers of the dialect of proto-Eastern Highlands Bantu that evolved into proto-Great Lakes Bantu had arrived in the western part of the Lakes region early in the last millennium B.C. After this first settlement, and based in part upon the successful incorporation of cereal agriculture and stock-raising, the proto-Great Lakes Bantu speech community broke up into five daughter groups: Luhyia, East Nyanza, Western Lakes, pre-RuGungu, and West Nyanza. The histories of the Western Lakes and West Nyanza groups occupy us in the final section. First, though, we must explore the growth of the new pastoral and agricultural knowledge and ability that propelled the expansion of Great Lakes Bantu speech.

Proto-Great Lakes Bantu-speakers lived between Rwenzori and northern Rwanda in the middle of the last millennium B.C. (Map 3B), and they were farmers. Much is already known about the food production system of their ancestors, the Eastern Highlands Bantu.[52] They were at home near the water's edge, and fishing was a major part of their food system.[53] Beside the shore, their descendants, the Great Lakes Bantu peoples, also settled the forest margins, where they planted a root-crop complex that included yams (*Dioscorea* sp.[54]), the oil palm (very likely *Elaeis guineensis*[55]), a type of bean and a type of gourd. They also grew the Bambara groundnut[56] and possibly the cowpea.[57] Though fishing and hunting certainly provided protein, oil, bone and skins for the first Great Lakes Bantu speakers who settled on the

[52] Ehret, 'Patterns', 35–9; *idem*, 'Agricultural history', 4ff; Schoenbrun, 'Early history', 165–7.

[53] For more on this well-known feature of the ancestral Bantu food system see Vansina, *Paths*, 88–92; Patrick R. Bennett, 'On the reconstruction of Bantu technology and its vocabulary', *Muntu*, III (1985), 121–35; Ehret, 'Lingusitic inferences', in Ehret and Posnansky (eds.), *The Archaeological*, 61–2.

[54] Yams cannot endure frost or waterlogged soils, and they thrive in warm, humid temperatures around 25–30 °C on deep, well-drained soils; I. C. Onweme, *The Tropical Tuber Crops* (New York, 1978), 12–13; J. W. Purseglove, *Tropical Crops : Monocotyledons* (London, 1972), 108–10.

[55] The oil palm will not tolerate frost either and thrives in more open, secondary forest or forest fringes where it may receive adequate sunlight. It will grow in riverine forests but not deep forests. It prefers a temperature range of 24–30 °C and an evenly distributed annual rainfall regime of about 2000 mm; Purseglove, *Monocotyledons*, 485–7; Jan Vansina 'Esquisse historique de l'agriculture en milieu forestier', *Muntu*, II (1985), 13.

[56] *Vigna subterranea* is a synonym for *Voandzeia s.* and is commonly intercropped in eastern Africa, doing well on dry, inferior soils up to 1500 meters above sea level. It prefers an evenly distributed rainfall of 600–1000 mm and a temperature range of 20–28 °C, FAO; *Traditional Food Plants*, 510–11.

[57] In eastern Africa, *Vigna unguiculata* tolerates droughts but not waterlogging, loves the sun, but is sensitive to cold, and frost kills it. It will stand a temperature range of 20–35 °C but prefers a range of 22–27 °C. It thrives at altitudes up to 2000 meters in a rainfall range of 280–4100 mm on many different soils but not alkaline. In the higher end

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Victoria Nyanza littoral, we should think of them, and of their ancestors the Eastern Highlands Bantu, as farmers par excéllence. The great range of words for crops and crop processing in the ancestral Great Lakes Bantu tradition alone reflects their deep involvement with arable agriculture.[58]

Early livestock production and trade

Though farmers, Bantu-speaking settlers in the west of the Great Lakes region raised small-stock as well; an Eastern Highlands Bantu breeding taxonomy for goats has been reconstructed by Ehret.[59] Beyond the importance of goats as a source of meat protein, farmers probably traded them and used them to pay for ritual or community services like funerals. Goats' browsing habits reduced woody vegetation and increased the extent of grassland.[60] Early herders may have employed goats thus, to control some tsetse and tick habitat, especially during the later periods in Great Lakes history as cattle-keeping became more important. Eastern Highlands Bantu peoples probably knew of cattle, as the few roots reconstructed so far suggest,[61] but the lack of a coherent breeding taxonomy implies that, although they may have owned and milked (7) a few animals, breeding itself remained unimportant to Eastern Highlands Bantu farmers before 500 B.C.

However, in the centuries before the turn of the Christian Era, Great Lakes Bantu farmers took a new interest in livestock. They coined new nouns for cattle-, goat- and sheep-breeding taxonomies, reflecting new levels of specificity achieved by their breeders. Even if Great Lakes peoples did not breed their own animals, they clearly understood how to do so, and what is more, they used them as sources of secondary products.

The exploitation of cattle for secondary products marked an important stage in the incorporation of livestock by Great Lakes farmers into their food systems. Milking and bleeding animals meant that, for the Great Lakes peoples, livestock were no longer seen solely as sources of meat and skins but came to be managed as renewable sources of nutrition. A range of new nouns and verbs were innovated for the processing of secondary products.[62] Great Lakes farmers churned fresh milk in a special calabash churn. They made sour milk and buttermilk in a milk pot, probably carved from wood. Cattle blood provided an iron supplement to an already nutritionally balanced diet. Dung from cows, possibly used as manure in certain nutrient-poor fields, took on enough significance in Great Lakes Bantu communities for them to create its own word.

Sherratt has already explained the importance of secondary products for

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of the rainfall range it is grown mainly for its leaves, with the seeds growing in importance to farmers in drier marginal lands. It is commonly intercropped with cereals and root crops; FAO, *Traditional Food Plants*, 518–22.

[58] The most comprehensive presentation of these data is Schoenbrun, 'Early history', 499–518.

[59] Christopher Ehret, 'Historical inferences from transformations in culture vocabularies', *Sprache und Geschichte in Afrika*, II (1980), 194.

[60] A. V. Oates, 'Goats as a possible weapon in the control of thornbush', *Rhodesian Agricultural Journal*, LIII (1956), 68–85.

[61] Ehret, 'Patterns', 39; Wrigley, 'Cattle', 250–62; Schoenbrun, 'Early history', 505–8.

[62] Linguistic data are available in Schoenbrun, *ibid.* 516–18.

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stimulating trade in the Old World.[63] And they may have done so in the Lakes region as well. The pastoral ways of life familiar from later periods in the history of Great Lakes polities such as Gisaka and Kitara certainly have their roots in this early development of expertise in milking and bleeding. I can reconstruct one term, *-*gaadU*, 'brown or red cattle color',[64] possibly to proto-Great Lakes, for a color that refers to cattle only. Detailed vocabularies for cattle colors, patterns or horn shape mark strongly a pastoral ideology, but they do not begin to turn up in Lakes Bantu language history until late in the first millennium A.D.[65]

Great Lakes farmers milked and bled their cattle probably for Cushitic and Sudanic markets as well as for internal consumption. Remarkably, and supporting this interpretation, lexical innovations pertaining to pastoralism diminish in number in Western Lakes and West Nyanza society. Livestock appear, then, to have been conduits for cross-cultural contacts during Great Lakes Bantu times, but cattle do not surface again in the linguistic record as important social or economic objects until after A.D. 1000.[66]

Indeed, Great Lakes Bantu were farmers, and the crops they planted, inherited partly from their Eastern Highlands ancestors, required specific environments in which to thrive. The rainfall and temperature requirements of yams and oil palms meant that communities depending largely on these crops would have lived at the forest fringes of the lower and wetter parts of the region (compare Maps 2, 3, and 4). The leguminous component tolerated lower humidity than the first two crops and contributed to soil fertility maintenance by fixing nitrogen. The gourd, of course, thrives in very dry climates as well as in wet tropical environments. But not too wet, for the intervening dry season best not exceed four months, nor should it be shorter than two if newly cleared yam fields are to be dry enough for planting.[67] And these drier zones could have been farmed as well by communities sowing cereals.

[63] Andrew Sherratt, 'The secondary exploitation of animals in the Old World', *World Archaeology*, XV (1983), 100.

[64] This root has the additional meaning 'large cow with long horns' in all members of the Rutara branch of West Nyanza in which there is a reflex. This might refer to the breed we call Zebu, being either *Bos indicus* or cross-breeds of *B. indicus* and *B. taurus*. Both of these breeds tend to be smaller than the small-humped Sanga breeds. Fiona Marshall now feels confident that either of the two former breeds was present in the eastern Rift by 2000 years ago; see F. Marshall, 'Rethinking the role of *Bos indicus* in Sub-Saharan Africa', *Current Anthropology*, XXX (1989), 236. The biological advantages of *B. indicus* in times and regions prone to environmental stress are well known. Its apparently late incorporation in the Great Lakes region lends support to the argument that risk minimization practices in the pastoral sector increased after A.D. 1000. For the latest classification of African cattle, see Caroline Grigson, 'An African origin for African cattle? Some archaeological evidence', *African Archaeological Review*, IX (1991), 119–44.

[65] See David L. Schoenbrun, 'Cattle herds and banana gardens', *African Archaeological Review*, forthcoming (1993).

[66] This revises views stated in Schoenbrun, 'Early history', 196, and is more in line with Schoenbrun, *ibid.* 227–34. For livestock in southern and western Uganda during the second millennium A.D. see Andrew Reid, 'The role of cattle in the Later Iron Age communities of Southern Uganda' (Ph.D. thesis, Cambridge University, 1991); and *idem*, 'Ntusi and its hinterland: further investigations of the Later Iron Age and pastoral ecology in Southern Uganda', *Nyame Akuma* XXXIII (1990), 26–8.

[67] Vansina, *Paths*, 86.

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Dry-land grains were early a part of Great Lakes fields. They inherited several root words for grain crops and processing from their Eastern Highlands ancestors, strongly suggesting that these cereals were part of Bantu crop rotations well before 500 B.C.[68] They harvested sorghum, *Pennisetum* and probably finger millet (*Eleusine*) and threshed, winnowed and ground these grains in wooden mortars. Several types of porridge and gruel also were made with grains.[69] Great Lakes Bantu communities themselves innovated three terms for cereals and their processing; a new word for 'sorghum flour', a verb 'to winnow' and a novel noun for *Eleusine* (2), borrowed from Central Sudanic. Indeed, Great Lakes Bantu-speaking groups developed grain agriculture in a multicultural context; several nouns and verbs in Great Lakes Bantu for different grains (2, 9, 15) and foods made from grains (1, 16, 17, 19) apparently have either Central or Eastern Sudanic origins. The Sudanians, not the Great Lakes Bantu, thus seem to have been the most important cereal-growers in the western Great Lakes region before A.D. 500.

The considerable expertise with cereals that Great Lakes farmers inherited may have been a driving force behind the break-up of Eastern Highlands Bantu into its daughter communities. With grain-growing abilities, families could more easily move to lower and drier areas at the edges of the Karagwe Depression, between Lake Rweru and Buhaya (Map 4B with Maps 2 and 3). Then, with livestock and grains in their generous crop complex, Great Lakes Bantu families possessed great advantages over their Sudanic and Cushitic neighbors as they moved eastward and settled the edges of the drier zones between the eastern-facing flanks of the western Rift and the high-rainfall zones along Victoria Nyanza's littoral. The agricultural achievements of Great Lakes Bantu-era farmers and herders formed the economic underpinnings of the Early Iron Age and foreshadowed the day when the Lakes region would be almost entirely Bantu-speaking.

In the middle of the last millennium B.C. an impressive collection of Africa's different linguistic groups thus practiced a considerable array of agricultural adaptations among the Great Lakes. The Eastern Highlands Bantu lived in the lower and wetter areas at the western fringe of our region, where their mixed grain- and root-cropping food system could thrive. Settled unevenly among these proto-Eastern Highlands Bantu-speaking farmers, the Central Sudanians were better adapted to farm and herd in the drier zones in the south and north of the Western Rift. Over the Rift edge, toward Victoria-Nyanza, the Eastern Sudanians lived in similar woody savanna areas in the northern Karagwe depression; their mixed pastoral and horticultural food systems being well-suited to the vagaries of rainfall in these micro-environments, undesirable to the newly arrived proto-Eastern Highlands Bantu. On the southern fringes of both Eastern Bantu and Central Sudanic settlements the Tale Southern Cushites herded. Their food producing expertise focused on raising, milking and bleeding cattle and on growing grains. Their emphasis on livestock lessened the competitive tension between them and their neighbors, the Bantu or Sudanic populations, for primary resources like land, water and pasture.

[68] See Ehret, 'Patterns', 38; Schoenbrun, 'Early history', 499–508.
[69] Language data are laid out in Schoenbrun, *ibid.* 509–13.

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The height of the Early Iron Age stretched across the millennium after 500 B.C. as Great Lakes speech spread within the Lakes region carrying with it the combined productive power of hitherto distinct food systems: fishing, trapping, bow hunting, gathering, root cropping, grain farming and both small stock and cattle raising. Thus, in the Early Iron Age, the growth of eclectic food systems throughout the Lakes region may have promoted, if not induced, the development of the advanced iron technology of the time. The demand for iron tools in newly active fields stimulated smelters and smiths to keep their furnaces burning. Equally important, but for future generations, the subsistence practices of Early Iron Age farmers and herders formed much of the productive groundwork for the developments of the next millennium, up to A.D. 1500, when Lakes communities began to form new social, economic and political identities with distinct regional textures. Those familiar developments flowed from the historical achievements of the Western Lakes and West Nyanza communities (Fig. 1 and Map 4); their stories form the last section of the paper.

WESTERN LAKES AND WEST NYANZA ACHIEVEMENTS: THE END OF THE EARLY IRON AGE, AFTER A.D. 500

The historical linguistic geography of the region proves that Great Lakes Bantu-speakers came to dominate virtually the entire area by A.D. 500. But precisely how much of that process involved their actual movement into unoccupied territories remains unclear.[70] Perhaps Great Lakes Bantu societies assimilated some non-Bantu-speaking groups, or some of the Sudanic and Cushitic people gradually adopted Bantu speech and culture.[71] A combination of these alternatives likely took place, and we should take care not to assume the migration of populations without demonstrating its occurrence and suggesting reasons for that movement. In the western half of the Great Lakes region relatively little territorial expansion seems to have taken place after the beginning of the Christian Era. The proto-Rwenzori-speakers were the exception; probably around A.D. 200 or 300 they spread northwards to the Rwenzori mountains. But, overall, peoples speaking one set of dialects, the proto-Western Lakes Bantu, settled in the highlands and well-watered reaches of the forest-fringe in and around the Western Rift not long after two thousand years ago (Map 4A). Their neighbors to the east spoke West Nyanza dialects and lived along and just to the west of the Victoria-Nyanza littoral, probably from western BuZinza to the Victoria Nile (Map 4B).

[70] In the drier grasslands between the Kagera and the Katonga, settlement was probably sparse if present at all. See Andrew Reid, 'The role of cattle'. But, in the lands between the lakeshore and the Kagera, south of its bend, occupation appears quite ancient. See Charles Nelson and Merrick Posnansky, 'The stone tools from the excavation of Nsongezi rock shelter', *Azania*, v (1970), 199–72; Peter R. Schmidt, 'Early Iron Age settlements and industrial locales in West Lake', *Tanzania Notes and Records*, LXXXIV/LXXXV (1980), 77–94.

[71] This is 'language shift'. For the emerging debate on this issue, see Derek Nurse, 'Language contact, creolization, and genetic linguistics: the case of Mwiini' (paper presented at the African Studies Annual Meeting, St Louis, MO, 1991); and S. G. Thomason and T. Kaufman, *Language Contact, Creolization, and Genetic Linguistics* (Berkeley, 1988).

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Like their distant Great Lakes Bantu ancestors, these people were pioneers. Following watercourses and lakeshores, they settled the uplands and wetter woody grasslands and broke out of the wet, low and humid forests. These early Lakes Bantu societies achieved their remarkably long period of local development and settlement in part by having developed a richly mixed agriculture. As part of the intense and shocking impact early iron production had on the forests in the territories of the fast-dissolving Great Lakes Bantu dialect chain, changes in farming, herding and land use drove the subsequently rapid development of the Western Lakes and West Nyanza communities in the earliest centuries A.D.

Western Lakes agriculture: more farming than herding

In the mountainous west, Western Lakes farmers responded innovatively to new pressures and opportunities in their farming systems after the turn of the Christian Era,[72] earlier developments toward agricultural complexity notwithstanding. In the corridor between Lakes Rweru and Kivu (Map 4A), these farmers learned of a new species of wild tuber from their Central Sudanic neighbors (14). But, the set of new terms for different types of fields perhaps tells most.

These farmers lived in a rapidly changing environment marked by shrinking forests and localized centers of iron production in the centuries before A.D. 500.[73] Several innovated terms for horticultural practice indicate a corresponding preoccupation with land clearance and field maintenance techniques. A new general term for 'field' and terms for 'to weed' and 'to open land by clearing trees' all portray their concern with the pressures of a changing physical environment. Rather than grapple with the integration of new crops, Western Lakes peoples apparently exploited a different sequence of field rotation. Different weeding techniques might have been part of shorter fallows or associated with new plant communities' successional behavior in an increasingly secondary forest. The same may be said of the new verb for clearing land. When added to an older term, **-tema* 'to clear by cutting', the new vocabulary echoed the new forest successions shaped by both earlier land clearance and demands for charcoal to smelt and work iron.

Western Lakes farmers cleared a wide variety of fields. By A.D. 200 or 300 Western Lakes farmers possessed a detailed vocabulary (at least nine terms[74]) for different types of fields, their preparation and their maintenance. Some of these were inherited from their very distant ancestors, the Eastern Highlands Bantu,[75] while still others were agricultural innovations undertaken in Great Lakes Bantu fields.[76]

[72] Archaeological and palynological evidence also exist. Van Grunderbeek *et al.*, provide fascinating though very thin evidence for a cereal-crop component in food systems of central Rwanda during the Early Iron Age; Van Grunderbeek *et al.*, *Le premier âge*, 27. And A. Gauthier produced the earliest direct evidence for cattle-keeping in the form of a single jaw with a *terminus ante quem* date of A.D. 300. See A. Gauthier, in Francis van Noten, *Histoire archéologique* (Butare and Tervuren, 1981), 32. The palynology is covered by Schoenbrun, 'The contours'.

[73] See Schoenbrun, 'Early history', ch. 3; Van Grunderbeek *et al.*, *Le premier âge*, 6–20.

[74] Schoenbrun, 'Early history', 500, 509–10, 535.

[75] Ehret, 'Agricultural history', 22; Schoenbrun, 'Early history', 500.

[76] Schoenbrun, *ibid.* 509–10.

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Using iron farming implements, developed during the last centuries B.C. in the Great Lakes Bantu era, Western Lakes farmers may have hastened the growth of field diversity. They inherited three ancient nouns for iron tools, 'iron hoe', 'worn-out iron hoe blade', and 'small iron blade for digging'.[77] Perhaps the new fields under sorghum that appeared in new areas contained heavier soils prone to water-logging in the rainy season that could be turned and planted much more easily with iron implements than with wooden digging sticks. After harvesting, these fields might have become available to livestock for stubble-grazing,[78] an easy way to green–manure fields. There may be, then, direct mutually reinforcing links between the exploitation of different kinds of local soils, the environmental degradation associated with industrial iron production (for example, in BuHaya[79]) and growing requirements in Western Lakes and West Nyanza villages for more and varied foods.

These different field types might also mirror the eclectic forms of crop-rotation inferred from palynological data. Different kinds of cultivated fields might plausibly be linked to growth in regional population densities. A growth in numbers in Western Lakes homesteads would have placed some pressure on cultivators, either to shorten the fallow length or to assume a wider range of subsistence poses in the form of different crop rotations. These processes might imply that Western Lakes farmers organized labor differently in order to increase surplus food production by multiplying their choices of crop rotations.

In Western Lakes homesteads livestock were less important than food crops. As farmers, they apparently found the rich soils and regular rains of the Kivu Rift a felicitous combination; they seem not to have added significantly to the cattle-keeping knowledge they had learned from their Eastern Sudanic, Central Sudanic and Tale Southern Cushitic neighbors. Still, Western Lakes societies developed new uses for their familiar domestic animals focused largely on small stock and milk production. The semantic reanalysis of the old Bantu term for 'goat' by the Western Lakes community produced a new meaning for **-búdI*, 'small livestock' in general. This meaning later shifted to 'sheep' in the Forest subgroup (Fig. 1). While the centuries around the turn of the Christian Era and nearing A.D. 500 saw a very diverse range of livestock-raising opportunities, Western Lakes homesteads prospered on a predominantly arable base until well into the second half of the first millennium A.D. Though beyond the scope of this paper, a second advance in the scale and importance of livestock began toward the end of the first millennium A.D., on the hills around the homesteads of farmers speaking Kivu languages.[80] This advance echoed one undertaken earlier and to the east of the Rift edge, in the pastures of West Nyanza farmers.

[77] Schoenbrun, *ibid.* 510–11.

[78] A development perhaps indirectly reflected in the palynological record from Kigezi. See especially Michael E. S. Morrison and Alan C. Hamilton, 'Vegetation and climate in the uplands of south-western Uganda during the later Pleistocene period, II: forest clearance and other vegetational changes in the Rukiga highlands during the last 8000 years', *Journal of Ecology*, LVI (1968), 26; Hamilton, *et al.*, 'Early forest clearance', 165, discussed in Schoenbrun, 'The contours'.

[79] Peter R. Schmidt and Donald Avery, 'Complex iron working and prehistoric culture in Tanzania', *Science*, CCI (1978), 1085–9.

[80] Schoenbrun, *ibid.* 133–6, 235–57.

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West Nyanza agriculture : farming and herding

In the districts just to the west of Victoria Nyanza (Map 4B), during the several centuries around A.D. 500, contact with neighbouring farmers and innovations in their own fields marked the West Nyanza development of a highly diversified farming base. New nouns for 'legume' and 'unripe legume' and for 'stiff millet porridge' point to ongoing innovations in the fields. West Nyanza farmers may have begun to intercrop beans with cereals and root-crops to exploit the nitrogen-fixing qualities legumes possess that make them so valuable for soil fertility maintenance. Beyond their value to healthy soils, West Nyanza families possessed in beans a valuable protein source to supplement their already rich cuisine. West Nyanza farmers also continued to learn about cereal-raising and processing from their Eastern Sudanic neighbors. Two loans (21, 22) show this. Perhaps the new noun for 'annually cultivated field' or 'garden' (21) embodied new practices such as planting cereals and tubers on a small scale, or a type of intercropping or crop rotation unfamiliar to West Nyanza farmers. These loanwords notwithstanding, the bulk of innovations in the West Nyanza farming vocabulary likely derived from Bantu roots;[81] they were worked out internally.

The extremely flexible range of soils in which *Vigna* spp. and *Sorghum* sp. could thrive furnished West Nyanza farmers with great choice in laying out different fields, whether intercropped or monocropped. Sorghum, which does well in heavier soils, may have been planted in purer stands in the black bottomlands where iron hoes were used to break and turn sod, while wooden digging sticks, certainly less expensive than iron hoes and produced at the homestead, would have been quite sufficient for preparing and maintaining intercropped fields of beans, tubers and (in certain rotations) some *Eleusine*. The Great Lakes Bantu innovations for iron tools all divulge the value to farmers of iron implements and suggest that they were recycled. And a new noun for 'wooden digging stick', in West Nyanza, attests to the latter's importance. The linguistic evidence may well be our only source of information on the use of wooden implements and the raising of root and runner crops. However, we must remember that fishing, gathering and hunting remained integral parts of food systems in the West Nyanza area and more widely throughout the Great Lakes region.

West Nyanzans developed their knowledge of cattle breeding largely on their own. Only a single loan in their language counters this trend, a generic term for 'cattle' (24 in Appendix). The original source for this was probably the Sog Eastern Sudanic mixed farmers resident in the Karagwe depression and near lake Kyoga, among whom lived the parts of the earliest West Nyanza communities to settle north of the mouth of the Kagera river (Map 4B). Though this item suggests an open attitude toward cattle in West Nyanza society, the rest of their pastoral practice recoverable by comparative linguistics diagnoses its internal elaboration.

The cow occupied a socially important role in West Nyanza society. A root **-gànà*, 'large herd of cattle', implies their need to discriminate among different size herds around their homesteads and suggests that cattle began to form important reservoirs of wealth, for at least some homesteads could

[81] Schoenbrun, 'Early history', 526–7.

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assemble large herds. Alternatively, West Nyanza herders might have observed and distinguished the larger herds of the adjacent Tale Southern Cushites or Sog Eastern Sudanians from their own small numbers of animals and, already possessing a perfectly good word for indicating a large number, they assigned *-*gànà* the meaning 'large herd' in addition to 'one hundred'. But, the West Nyanza root *-*sumba*, 'young herder', means that in some of their pastures young males practiced herd management. The fact that the term's wider semantic field in Eastern Bantu includes a (junior) male designation of some sort ('suitor' in KiSwahili, 'young man' in Luhyia) lends strong support to the view that cattle began to emerge as markers of stratification within and between West Nyanza homesteads by A.D. 500.

These few items only point to age- and gender-specific divisions of labor in the care of domestic animals. But taken together with innovations in West Nyanza farming already discussed and added to the large body of knowledge inherited from Great Lakes times, the full West Nyanza complement of vocabulary for crops, tools, field types, domestic animals, their breeding and their secondary products strongly suggests a division of labor tied to the intensification of arable and pastoral food production.

Even though Western Lakes peoples focused clearly on farming rather than herding, both they and West Nyanza communities grazed and bred cattle more than 2000 years ago. They milked them as well. The new terms for 'fresh milk', innovated in both communities, support this conclusion. It seems that after the initial contact between Great Lakes farmers and their more pastoral Sudanic- and Cushitic-speaking neighbors, livestock rose to importance earlier in West Nyanza pastures than around Western Lakes homesteads. In the case of West Nyanza, pastoralism was largely internally developed, and we can see in the linguistic evidence the beginnings of new divisions of labor that hint at correspondingly different categories of social identity as cattle came to signal family wealth.

Root crops, grain crops and a familiarity with cattle breeding practices provided the Western Lakes- and West Nyanza-speaking peoples with the economic tools to confront a series of vegetational changes initiated by their ancestors before the turn of the Christian era. This same agricultural adaptation, elaborated and firmly established along the western Victoria littoral and in the Kivu Rift, rather quickly differentiated itself between the lowland herders and farmers (West Nyanza-speakers) and Western Lakes-speaking highland farmers and herders (Maps 4A and 4B).

The large Western Lakes and West Nyanza communities managed the combination of changing ecological systems, iron production and socio-cultural assimilation that eclipsed both the Eastern and Central Sudanic and the Tale Southern Cushitic roles in the Great Lakes region as A.D. 1000 approached. The paper only tells this one side of the story. We do not know what the non-Bantu peoples learned from each other and their Bantu-speaking neighbors. Nevertheless, we can see that those non-Bantu peoples played a pivotal role in the adoption of cattle-keeping and grain farming by several of the Great Lakes Bantu communities, and a dry-land mixed agriculture in turn was to become a contributing factor to the further spread of Bantu tongue in later centuries.

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CONCLUSION

The roots of the modern social, economic, environmental and political formations in the Lakes region lie in the earliest periods of Lakes history, even before the advent of iron and pottery. Developed and diverse food systems preceded the emergence of the Early Iron Age but flowered during the roughly thousand years after 500 B.C. The varied cultural contexts matched the diversity of this florescence. People speaking several different languages were herding cattle, sowing grains, planting root crops, fishing and hunting all across the area. No fewer than three such groups lived in the central grasslands near the Kagera bend; archaeologists should search there for early evidence of cattle-keeping and cereal raising.

In the earliest periods, beginning well before 3000 years ago, the first food producers settled in the western parts of the region. Their food systems, different enough in their requirements of rainfall and soils, produced a sort of patchwork pattern of settlement. Mixed-farming Sog Eastern Sudanic peoples and the primarily pastoral Tale Southern Cushites settled side by side between the Victoria Nyanza littoral and the Western Rift highlands. The mainly root-cropping and fishing Eastern Highlands Bantu and the cereal and stock raising Central Sudanic peoples settled amidst one other, divided primarily by the high moisture and warm temperature requirements of the Bantu crop complex. These farmers may have been partly responsible for the earliest constrictions of forests in the western Lakes region. Their technology may have been neolithic, but iron before 500 B.C. cannot be ruled out.

The environmental specificity of this settlement pattern led to long-standing interaction among the various agricultural traditions. Agricultural practices also shaped and responded to environmental changes such as receding forest cover and shrinking rainfall amounts, respectively. The Eastern Highlands Bantu took up cereal farming and one of their descendant groups, the Great Lakes Bantu, later developed large stock-raising, presumably in order to expand their settlement options and to increase the productivity of their food systems. They probably learned about the new food ways through intermarriage, cooperative hunting parties and trade relations linking the products unique to each environmental region and historical tradition. For example, Eastern Highlands Bantu peoples may have exchanged dried fish and yam flour for sheep or cattle with the Sudanic communities. Or Tale Southern Cushites may have given bulls to Great Lakes Bantu peoples in return for iron implements.

The story told in this paper employs large-scale environmental, demographic and technological explanations of historical change, resting as it does on a narrow range of linguistic evidence in these semantic fields. But more than instrumental processes lay at the heart of the Eastern Bantu agricultural synthesis. If further study on the historical transformations of Bantu kinship systems bears anticipated fruit,[82] we may soon be in a position to offer sociological hypotheses to explain the apparent dissolution into Great Lakes

[82] See Vansina, *Paths*, 73–83, 152–5; and see the important critique of Vansina in Christine Ahmed, 'Not from a rib: the use of gender and gender dynamics to unlock early African history' (paper presented at the African Studies Association meeting, St Louis, 1991).

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Bantu societies of Sudanic and Cushitic ones. The kernel of the argument resides in Eastern Highlands Bantu society having been a matrilineal and a matrilocal one, wherein women superintended and carried out agricultural production and outsider males were easily incorporated. Two implications follow from this formulation. Either non-Bantu men married into Eastern Highlands Bantu and, later, Great Lakes Bantu families in order to join their productive expertise to their new wife's and her sisters' (also ensuring that their children would come to speak as a first language a Bantu tongue).[83] Or Bantu-speaking women married into Sudanic and Cushitic societies, maintained strong ties to their natal homesteads and raised bilingual children.[84] A number of as yet unsubstantiated assumptions leave these explanations of the rise to prominence of Great Lakes Bantu-speaking society no more than speculative. We do not yet know for certain if Eastern Highlands Bantu society was, indeed, matrilineal. And, if it was, had women control over all the main features of agricultural practice? Furthermore, it seems from linguistic and ethnographic evidence that, at least in Great Lakes Bantu societies, during the period of the Early Iron Age and the early florescence of Bantu-speakers as cattle-users, their kinship system was already becoming patrilineal, as are nearly all modern Lakes societies. For now, though, these suggestions must remain tentative.

At all events, probably multilingual groups that increasingly spoke Great Lakes Bantu as a first language rose to prominence in this multi-cultural context. The boundaries enclosing language, food system and settlement location began to break down after 500 B.C. The cumulative agricultural expertise resulting from centuries of cultural interaction (and preserved in modern Lakes Bantu vocabularies) was a prerequisite for the ascendancy of Bantu speech. Several attributes described this dominance, more than 2000 years ago. Some Great Lakes Bantu societies exploited the cattle-keeping and grain-producing parts of their food systems in order to expand into the drier zones of the region. As responses to environmental changes they themselves had induced, partly through their metallurgical and agricultural abilities, this expansion ultimately assimilated Cushitic and Sudanic society. But that process did not draw to a close until after A.D. 1000.

The 500 to 700 years after 500 B.C. marked the heyday of the Early Iron Age, during which Great Lakes Bantu communities developed their economic power and left a lasting impression on the landscape. Though food, environment and technology were closely linked during that time, I believe that the demands on land, labor and technical expertise made by the farmers and herders who undertook the agricultural developments, charted here as a blending of historical traditions, characterize more appropriately a period in the African past too often linked only to technology. The Early Iron Age in the Lakes region was as much an age of cultural contact and economic development as it was a period of metallurgical advance.

In the twilight of the Early Iron Age, between A.D 300 and A.D. 800, Western Lakes and West Nyanza Bantu farmers and herders elaborated on the achievements of their Great Lakes Bantu predecessors. Long-standing farming practices and the shocking environmental impact of iron production may have reduced soil fertility and increased erosion, thus making quality lands a relatively rare item during the end of the Early Iron Age. To respond

[83] Ahmed, 'Not from a rib', 7–8. [84] Schoenbrun, 'Early history', 293.

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to this, Western Lakes and West Nyanza farmers may have shortened their fallows in existing, healthy fields either to provide more regional trade items or to offset pressures from the probably robust demographic expansion that accompanied the breakdown of the old agroecological boundaries.[85] Some West Nyanza and Western Lakes farmers experimented with bananas, and some West Nyanza peoples refined their pastoral social values.[86] Indeed, in this period some of the basic attributes of modern Lakes regional identities began to form; densely sedentary lakeshore and riverine root-cropping and fishing societies interlinked with more mobile and more pastoral societies. These identities, and the highly productive, because broadly-based food systems in which they were rooted firmly, constitute two vital arenas for the historical struggles in the era of political and social change after A.D. 800.

SUMMARY

A history of food systems in Africa's Great Lakes region is presented using mostly historical linguistic sources, with help from archaeology and paleoecology. The paper moves beyond understanding the causes and consequences of iron-working as the most important feature of the period between *c.* 1000 B.C. and *c.* A.D. 500. I argue that a history of agriculture both gives context to changes in technology and introduces powerful new explanations for historical processes of settlement and occupational specialization that took place.

Between 1000 B.C. and 500 B.C., in the Great Lakes region, speakers of three of Africa's four major language families practiced distinguishable food-producing systems. Two groups, Central Sudanian and Sog Eastern Sudanian, depended mainly on growing cereals and raising livestock for their sustenance. The third group, the Tale Southern Cushites, gave decidedly greater emphasis to cattle but probably also grew grains. A fourth group, the Great Lakes Bantu, grew root crops, fished and raised cattle and grain. They inherited much of their knowledge of these techniques, other than cattle-raising, from earlier Eastern Highlands Bantu-speakers. But they incorporated cattle and some grains through long-standing contacts with the two Sudanian and the Southern Cushitic communities. The eclectic food system they thus created allowed them to carry their unified, complex food-producing system throughout the wide variety of environments that they encountered in the Lakes region. After *c.* A.D. 200 descendants of the Great Lakes Bantu refined this synthesis; they emphasized livestock raising inland from Lake Victoria, and mixed farmers spread throughout the Kivu Rift. Technological, demographic, ecological and sociological explanations of the technological evidence are offered.

APPENDIX. LOANWORDS FOR FOOD PRODUCTION IN LAKES BANTU

Central Sudanic loans in Eastern Bantu

(1)	*-(i)gi	'gruel'	PCSud *Gi 'sorghum'; Moru-Madi ego 'porridge'.
(2)	*-le	'millet, *Eleusine*'	PCSud re 'seed'.

[85] This may well have been the context for the initial integration of the constellation of symbols and ritual process in the *umuganuro* 'first fruits' ceremonies that are shared by enough Western Lakes societies to suggest that *umuganuro* has very ancient roots. See Newbury, *Kings and Clans*, 200–26, 290 n. 16, 291 n. 23.

[86] The data for these developments will appear in Schoenbrun, 'Cattle herds'.

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(3)	*-nU	'wooden mortar'	CSud root.[87]
(4)	*-ba	'enclosure or yard'	PCSud *Ba 'residence, homestead, village'.

Central Sudanic Loans in Great Lakes Bantu

(5)	*-shenyi	'sand'	PCSud *tsenya Moru-Madi *tsényá 'sand'; Moru senye 'sand'.
(6)	*-pio	'herd'	PCSud *pi- 'to increase'. plus -V deverb. suffix.
(7)	*-aandI	'wooden milking jug'	PCSud: Madi ndre 'to milk'.
(8)	*-pu	'skin'	PCSud *pu 'to remove'; Baledha meaning 'to skin' Or PCSud *pO 'to tear off'; Moru-Madi: Lugbara po 'to peel'; Mangbetu -apuO 'to husk'. Rutara; LuSoga; Western Lakes; EkeGusII erio 'removed skin'.
(9)	*-lo	'finger millet'	CSud: *Do, Lugbara Dö 'finger millet'.

Central Sudanic loans in West Nyanza

(10)	*kaiga	'unripe bean'	PCSud *ga or *Ga 'hard lump, etc.,'. ekikagga in LuGanda; ekikaiga in Eki-Haya.
(11)	*-bo	'tick'	PCSud: Lugbara obu 'jigger or burrowing larva'. (The root is *-bu) Full distribution unknown.
(12)	*-bubi	'dappled'	PCSud; Mangbetu obubu 'spotted'.

Central Sudanic loans in Western Lakes

(13)	*-nzu	'gruel' (general term)	PCSud: *nZu 'to ooze'. K-A: Kresh g-undju 'thick (liquid)'.
(14)	*-liga	'yam sp.'	CSud: Mangbetu n-adigá 'a kind of yam'.

Sog Eastern Sudanic loans in Eastern Bantu

(15)	*-βèlé	*Pennisetum*	ESud *bel or *Bel 'grain' WNilotic 'sorgum'.[88]
(16)	*-kIma	'gruel' (general)	ESud: PEnNilotic *kima PKal *kim- 'porridge'.
(17)	*-gadi	'gruel'	Ultimately from ESud *Kal 'porridge', but widely distributed in EnBantu; comparatively recent loan spread.

Sog Eastern Sudanic loans in Great Lakes Bantu

(18)	*-ka	'head of cattle'	NiloSah *-ka 'homestead'. Songay reflex: ga 'cattle camp'. Unlikely alternative derivation: PSoCush: *xah- 'body, skin, flesh'.

[87] See Ehret, 'The African Great Lakes', 6, and Schoenbrun, 'Early history', 503.
[88] See Ehret, 'The African Great Lakes', 9, and Schoenbrun, 'Early history', 502.

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(19)	*-sela *-sala	'porridge' or 'millet flour'	Kir-Abbaian: Daju *seR- 'porridge'. Luhyia, Rutara, and East Nyanza. Also in Rwenzori *-saara 'porridge', LuGanda -sera and GiKuria -sara '*Eleusine* flour'.

Sog Eastern Sudanic Loans in West Nyanza Bantu

(20)	*-gele	'foot'	NSah: Kir *k'ɛɛL, 'foot'.
(21)	*-sili	'cultivated field'	PKuliak: Soo: sed.
(22)	*-tele	'flour type'.	Kir-Abbaian: Surma: Murle tɛlɛ 'porridge'; Gaam, etc.
(23)	-sozi	'mountain'.	Kuliak: *sog- 'hill, stone'. The presumed West Nyanza reconstruction is *-sogI.
(24)	*-te	'head of cattle'	ESud source indicated here, rather than CSud which had *ti.[89]

Tale Southern Cushitic loans in Great Lakes Bantu

(25)	*-kaapU, *-kaaβU	'head of cattle'	PRift: Burungi *Ka?afu 'gateway to homestead or pen'.
(26)	*-masa	'bull or cow which has not produced young'	Kw'adza mahas-iko 'ox' from PSCush: *mah-$^{-}$, 'to be sterile' plus *s agent noun suffix.
(27)	*-saI	'cow dung'	Cf. PRift: Iraqw tseHa 'dry cow dung' (root *ts'eH-)?
(28)	*-tale	'lion'	PSCush *taal-$^{+}$ 'lioness'.
(29)	*-punu	'(wild) pig'	PWRift *bayn- 'pig' (from *p'∧n- or *-b∩n). In KiHa; RuKoonzo; EkiHaya, LuNyoro ('pig'), RuCiga, RunyaNkore; RuGungu. A loan in SiSumbwa.
(30)	*-saato	'animal skin'	SoCush root seen in PRift *ch'at 'hide tether'.

[89] See Wrigley, 'Cattle and language', 250–3 and 256, where he comes close to the position taken here. I employ a different etymology than does André Coupez, 'Linguistic taboo concerning cattle among the interlacustrine Bantu', *Acts of the African Languages Congress* (Pretoria, 1976), 226–7.

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